

Chapter 6: Light and Space

Comprehensive Study Notes & Worksheet

Cambridge Stage 8 Science

Name: _____

Date: _____

6.1 Reflection of Light

Light travels in extremely fast, _____ lines. When light hits a surface and bounces off, we call this _____.

- **The Law of Reflection:** The angle of _____ (i) is ALWAYS equal to the angle of _____ (r).

- $i = r$

- **Key Terms for Ray Diagrams:**

- _____: The incoming ray of light hitting the mirror.
- _____: The ray of light bouncing off the mirror.
- _____: An imaginary dotted line drawn at exactly _____° (a right angle) to the mirror's surface at the exact point where the light hits.
- **Angles:** Angles are ALWAYS measured between the light ray and the _____ (never between the ray and the mirror itself!).

- **Types of Reflection:**

- _____ **Reflection:** Happens on perfectly smooth, shiny surfaces (like a mirror or still water). All light rays reflect in the same direction, creating a clear image.
- _____ **Reflection:** Happens on rough surfaces (like paper or wood). Light rays scatter in many different directions, so no clear image is formed.

- **Images in a Plane (Flat) Mirror:**

- They are _____ (the light rays don't actually meet behind the mirror, so the image cannot be projected onto a screen).
- They are exactly the same size as the object.
- They appear to be the same distance *behind* the mirror as the object is in front.
- They are _____ (left and right are swapped—this is why "AMBU-LANCE" is written backwards on the front of the vehicle!).

6.2 Refraction of Light

_____ is the bending of light when it passes from one transparent material (medium) into another.

- **Why does it happen?** Light travels at different _____ in different materials depending on their _____. It travels fastest in a vacuum, very fast in air, slower in water, and even slower in glass. When light hits a new medium at an angle, the change in speed causes the light ray to bend.
- **The Rules of Bending:**
 - **Less dense to more dense (e.g., Air into Glass/Water):** The light slows down and bends _____ the normal.
 - **More dense to less dense (e.g., Glass/Water into Air):** The light speeds up and bends _____ the normal.
- **Real-world examples of Refraction:**
 - **The Broken Pencil:** A straight pencil placed in a glass of water looks _____ or broken at the surface because the light traveling from the underwater part of the pencil changes direction as it enters the air.
 - **Apparent Depth:** Swimming pools look _____ than they actually are. If you look at a coin at the bottom of a pool, refraction makes the coin appear higher up than its true position.
 - **Lenses:** Glasses, microscopes, and telescopes use curved pieces of glass (lenses) to deliberately refract light to magnify objects or correct vision.

6.3 Making Rainbows (Dispersion) & Seeing Colours

White light from the Sun isn't actually white—it's a _____ of all the colours of the rainbow combined!

- _____: The splitting of white light into a spectrum of colours. We can do this using a triangular glass _____.
- **The Spectrum:** The colours of white light always appear in this exact order: **Red, Orange, Yellow, Green, Blue, Indigo, Violet** (ROYGBIV).
- **How it works:**
 - Different colours of light travel at slightly different speeds inside glass, meaning they refract (bend) at slightly different angles.
 - **Red light** bends the _____ (it is the fastest in glass).
 - **Violet light** bends the _____ (it is the slowest in glass).
- **Nature's Prisms:** Real rainbows form when millions of tiny water droplets suspended in the air after a rainstorm act as tiny prisms, refracting and dispersing the sunlight.
- **Why are objects coloured? (Absorption and Reflection):**
 - When white light hits a coloured object, the object _____ some colours and _____ others.
 - A red apple looks red because it reflects red light into our eyes and absorbs all the other colours.
 - A perfectly _____ object reflects *all* colours.
 - A perfectly _____ object absorbs *all* colours (which is why wearing black on a sunny day makes you feel hotter!).

6.4 & 6.5 Galaxies and the Universe

Moving from light waves to the vastness of space!

- **What is a Galaxy?** A massive, spinning collection of billions of stars, dust, gas, and dark matter, all held together by _____.
- **Types of Galaxies:**
 - _____ **Galaxies:** Have a central bulge and twisting arms made of stars and gas (e.g., our _____).
 - _____ **Galaxies:** Shaped like giant ovals or spheres. They mostly contain older stars and have less dust and gas.
 - _____ **Galaxies:** Have no defined shape, often because they have collided with other galaxies.
- **The Milky Way:** This is our home galaxy. Our Solar System is located on one of its outer spiral arms, called the Orion Arm. If you look at the night sky in a very dark place, you can see the dense center of our galaxy as a milky band of light.
- **The Universe:** The universe is everything that exists. It contains billions of different galaxies separated by vast amounts of completely empty space (a vacuum). Scientists believe it began expanding from a single point during the _____ around 13.8 billion years ago.
- **Light Years:** Because space is so enormous, measuring in kilometers is impossible (the numbers would be too long). Scientists use _____ to measure distance.
 - A light-year is the _____ that light travels in one Earth year (about 9.46 trillion kilometers).
 - *Fun Fact:* The nearest star to us (other than the Sun) is Proxima Centauri, which is about 4.2 light-years away. That means the light we see from it today actually left the star 4.2 years ago!

6.6 Rocks in Space

Not everything in space is a huge star or a round planet. There is a lot of rocky and icy debris leftover from the formation of the solar system 4.6 billion years ago.

- _____: Lumps of solid rock and metal.
 - They have irregular, bumpy shapes.
 - Most are found in the _____, a huge ring of space rubble orbiting the Sun between the orbits of _____ and _____.
 - *Fun fact:* It is widely believed that a massive asteroid impact on Earth 66 million years ago (the Chicxulub crater) caused the extinction of the dinosaurs.
- _____: Often called "dirty snowballs," these are made of loose rock, dust, and frozen gases/ice.
 - They have highly oval-shaped (elliptical) orbits that take them to the very freezing edges of the solar system and then close to the Sun.
 - As they get close to the heat of the Sun, the ice melts and vaporises into gas. This forms a glowing _____ (cloud) around it and a long _____ that can stretch for millions of kilometers.
 - The tail *always* points _____ from the Sun because it is pushed by solar winds.
 - *Famous Example:* Halley's Comet, which is visible from Earth every 75–76 years.
- **Meteoroids, Meteors, and Meteorites:** (The 3 M's of Space Rocks)
 - _____: A small piece of rock or metal travelling completely freely through space.
 - _____: A meteoroid that enters Earth's atmosphere. It travels so fast that air friction heats it up, causing it to burn and create a bright streak of light. We commonly call these "**shooting stars**".
 - _____: If a meteor is large enough, it won't burn up completely. The chunk of rock that actually survives and crashes into the Earth's surface is called a meteorite.

Chapter 6: Student Worksheet

Read the study notes carefully, then answer all 50 questions below.

Section A: Multiple Choice Questions (Circle the correct answer)

1. How does light travel?
 - A. In wavy lines
 - B. In straight lines
 - C. In circles
 - D. It teleports
2. According to the Law of Reflection, if the angle of incidence is 40° , what is the angle of reflection?
 - A. 20°
 - B. 50°
 - C. 40°
 - D. 90°
3. Where is the angle of incidence measured?
 - A. Between the incident ray and the mirror
 - B. Between the incident ray and the normal
 - C. Between the reflected ray and the normal
 - D. Between the reflected ray and the mirror
4. Which type of reflection happens on a completely smooth, shiny surface like still water?
 - A. Diffuse reflection
 - B. Virtual reflection
 - C. Specular reflection
 - D. Refractive reflection
5. Why does the word "AMBULANCE" appear backwards on emergency vehicles?
 - A. Due to diffuse reflection
 - B. Due to refraction in the air
 - C. Due to lateral inversion in plane mirrors
 - D. So the driver can read it
6. What happens to light when it travels from air into a glass block?
 - A. It speeds up and bends towards the normal
 - B. It speeds up and bends away from the normal
 - C. It slows down and bends towards the normal
 - D. It slows down and bends away from the normal

7. Which phenomenon makes a swimming pool look shallower than it actually is?
 - A. Reflection
 - B. Refraction
 - C. Dispersion
 - D. Absorption
8. What equipment is used to split white light into a spectrum in a laboratory?
 - A. A plane mirror
 - B. A glass block
 - C. A triangular glass prism
 - D. A magnifying lens
9. Which colour of light travels the SLOWEST inside a glass prism?
 - A. Red
 - B. Green
 - C. Yellow
 - D. Violet
10. Why does a green leaf appear green under sunlight?
 - A. It absorbs green light and reflects all other colours.
 - B. It reflects green light and absorbs all other colours.
 - C. It creates its own green light.
 - D. It refracts green light away from the eye.
11. Which shape accurately describes our home galaxy, the Milky Way?
 - A. Elliptical
 - B. Irregular
 - C. Spiral
 - D. Spherical
12. What does a "light-year" measure?
 - A. The speed of a star
 - B. Time
 - C. The distance light travels in one year
 - D. The brightness of a galaxy
13. Where is the main Asteroid Belt located?
 - A. Between Earth and Mars
 - B. Between Mars and Jupiter
 - C. Past the orbit of Neptune
 - D. Between Venus and Earth

14. Why does a comet have a glowing tail?
- A. It is burning up in Earth's atmosphere.
 - B. The ice melts and vaporises into gas near the Sun.
 - C. It reflects the light of nearby stars.
 - D. It is made of fire.
15. What do we call a piece of space rock that survives the atmosphere and hits Earth's surface?
- A. Meteoroid
 - B. Meteor
 - C. Meteorite
 - D. Asteroid

Section B: Fill in the Blanks

1. When light hits a surface and bounces off, we call this _____.
2. The imaginary dotted line drawn at exactly 90° to a mirror's surface is called the _____.
3. _____ reflection happens on rough surfaces like wood, scattering light in many directions.
4. An image in a plane mirror cannot be projected onto a screen because it is a _____ image.
5. The bending of light when it passes from one transparent medium to another is called _____.
6. Light travels fastest in a _____.
7. When light travels from water into air (more dense to less dense), it bends _____ from the normal.
8. The splitting of white light into a spectrum of colours is known as _____.
9. The colours of the visible spectrum in order are Red, Orange, Yellow, Green, Blue, Indigo, and _____.
10. A perfectly _____ object absorbs all colours, which is why it heats up faster on a sunny day.
11. A _____ is a massive collection of billions of stars, dust, and gas held together by gravity.
12. Scientists believe the universe began expanding from a single point during the _____.
13. Comets are often described as "dirty _____" because they are made of loose rock, dust, and ice.
14. A comet's tail always points _____ from the Sun because it is pushed by solar winds.
15. The common name for a meteor burning up in the Earth's atmosphere is a "_____ star".

Section C: Short Answer & Explanation

1. State the Law of Reflection.

2. Explain the difference between the incident ray and the reflected ray.

3. List three properties of an image formed in a plane (flat) mirror.

4. Explain clearly why a straight pencil looks bent or broken when placed in a glass of water.

5. Explain why a violet light ray bends more than a red light ray when passing through a glass prism.

6. How do real rainbows form in nature after a rainstorm?

7. If you wear a white t-shirt on a sunny day, why do you feel cooler than someone wearing a black t-shirt?

8. Name the three main shapes/types of galaxies found in the universe.

9. Why do scientists use "light-years" to measure distances in space instead of kilometers?

10. Proxima Centauri is 4.2 light-years away. If that star exploded today, when would we see it happen on Earth? Explain.

11. What is the difference between an asteroid and a comet in terms of what they are made of?

12. Describe the shape of a comet's orbit around the Sun.

13. Define what a meteoroid is.

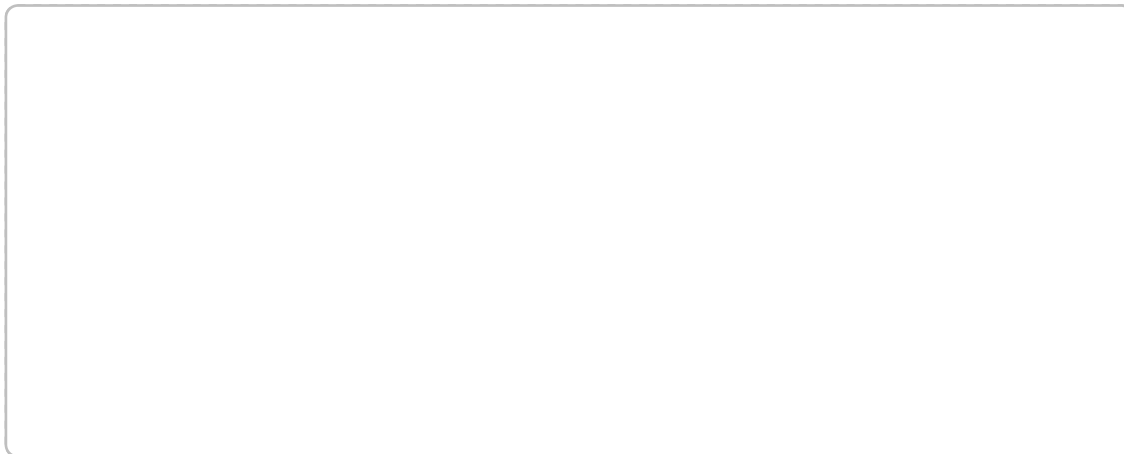
14. Why does a meteor create a bright streak of light when it enters Earth's atmosphere?

15. What is the scientific theory regarding the extinction of the dinosaurs 66 million years ago?

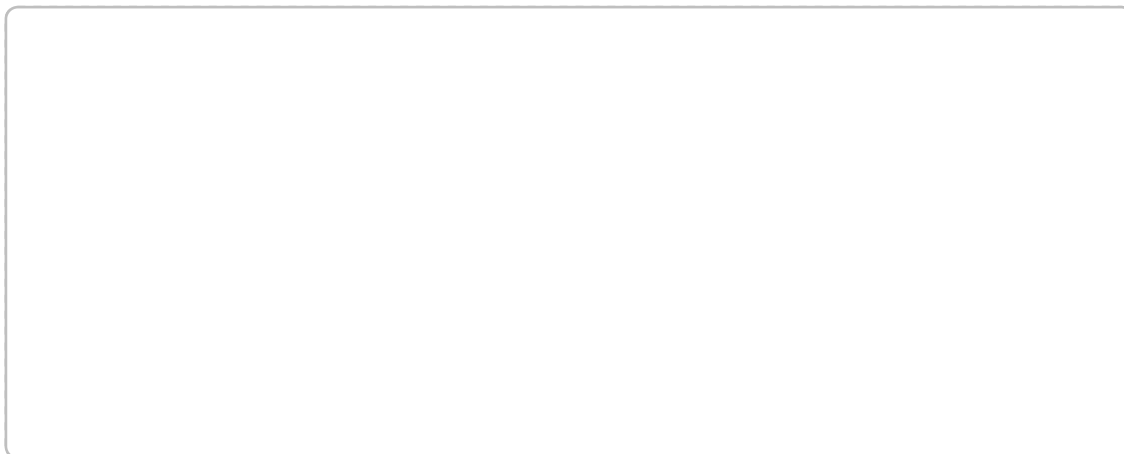
Section D: Drawing & Sketching Challenges

Use a pencil and ruler where necessary to complete the following diagrams based on your notes.

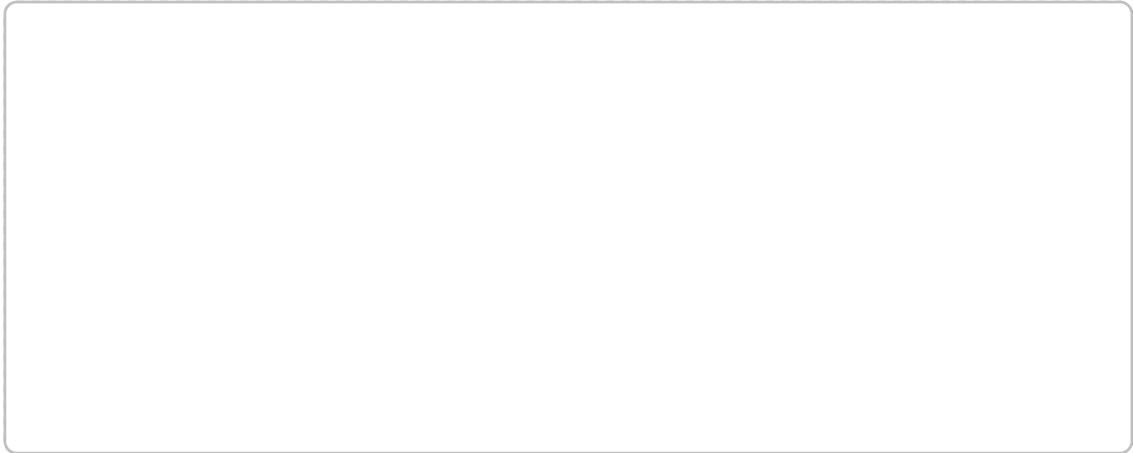
1. **Reflection Diagram:** Sketch a flat mirror. Draw an incident ray hitting the mirror, the normal line, and the reflected ray. Label the angle of incidence (i) and the angle of reflection (r).



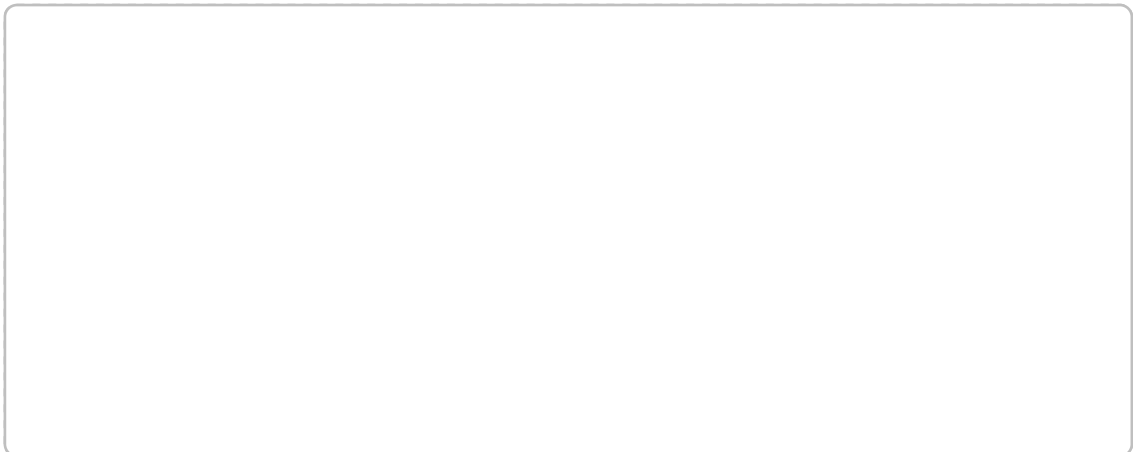
2. **Refraction Diagram:** Draw a rectangular glass block. Show a ray of light traveling from the air, entering the glass block, and then exiting back into the air. Show how the light bends at both boundaries.



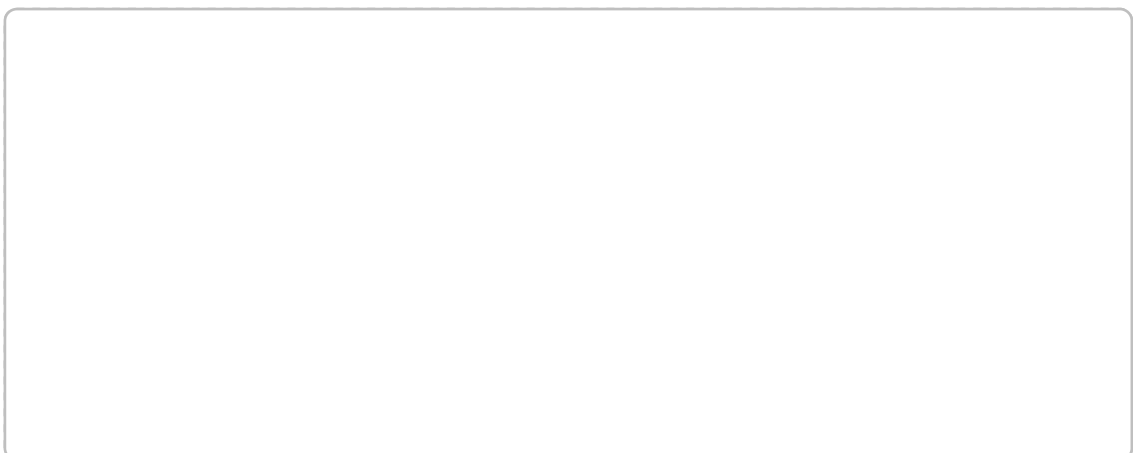
3. **Dispersion Diagram:** Draw a triangular glass prism. Show white light entering the prism and splitting. Label the colours that bend the **most** and the **least** as they exit.



4. **Galaxy Sketch:** Sketch the shape of a **spiral galaxy** (like our Milky Way) viewed from above. Place an 'X' to mark the approximate location of our Solar System on one of the outer arms.



5. **Comet Orbit:** Sketch the Sun in the center. Draw the highly elliptical orbit of a comet around the Sun. Draw the comet itself at a point close to the Sun, ensuring you draw the tail pointing in the correct direction.



End of Worksheet. Great job!